GADDAM HRUSHIKESH RAJ SAI

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.28
S.R. Junior College, Karimnagar	Class XII (Telangana State Board of Secondary Education)	2019	98.20%
Sai Grace Grammar School, Siddipet	Class X (Board of Secondary Education, Telangana)	2017	10.00%

SKILLS

Programming: C++, C, Python, Bash

Systems & Networking: Linux, TCP/IP, KVM/libvirt, OpenStack, Proxmox, VLANs, Linux Bridges, Routing/NAT

Tools: Git, Azure DevOps

PROJECTS

OpenStack Private Cloud | Capstone Project

- **Developing a multi-node OpenStack environment on Linux/KVM**, isolating control-plane services in VMs and separating control, data and storage traffic across dedicated network planes.
- **Integrated core control-plane and compute services** through Keystone, Glance, Placement and Nova, validating resource discovery, scheduling and VM lifecycle flows.
- **Engineering Neutron/OVN tenant networking**, debugging VLANs, Linux bridges, routing/NAT and packet paths while bringing up gateway/compute connectivity and metadata services.

Cryptanalysis of Reduced-Round SNOW-V/SNOW-Vi | Topics in Cryptanalysis

- **Worked on a parallel C++ implementation of reduced-round SNOW-V/SNOW-Vi cryptanalysis**, using AES-NI, OpenMP and 256-core HPC to reproduce and validate published attack parameters.
- **Contributed to a staged search across 16,384 differential pairs**, replacing sparse random sampling with progressive filtering and reducing estimated key-recovery complexity from **2^233.99 to 2^128**.

4-Round Custom DES Key Recovery using Differential Cryptanalysis | Cryptology

- **Implemented a complete differential key-recovery pipeline for custom 4-round DES**, using S-box DDT analysis and chosen-plaintext differentials to recover 42 round-key bits, then reversing the key schedule and searching the remaining 14 bits.
- **Built a local validation workflow before using the constrained oracle**, testing against known keys until the attack was reproducible and recovering the final 56-bit key with **at most 10 plaintext-ciphertext pairs**.

Proxmox VE Cluster & VM Automation

- **Deployed Proxmox VE across bare-metal and nested KVM hosts**, building reusable Cloud-Init/QEMU Guest Agent templates and validating cloned VM bring-up in **5-15 seconds**.
- **Automated provisioning and VM lifecycle operations through REST APIs**, using API tokens and least-privilege RBAC for controlled, delegated access.

WORK EXPERIENCE

Software Development Engineer I -- MAQ Software | Data Engineering & Analytics

- **Contributed to a production data-pipeline redesign**, modifying ADF orchestration, ADLS staging paths and Databricks processing to reduce refresh time from **2 days to 10 hours**.
- **Diagnosed production failures across a multi-stage data pipeline**, tracing issues across SQL, Databricks, ADLS and upstream systems and validating data freshness and consistency at each layer.
- **Implemented and validated cross-layer migration/configuration changes** across ADF, ADLS and Databricks, updating dependent pipelines/notebooks and coordinating builds and releases through Azure DevOps.

RELEVANT COURSES

Advanced Data Structures & Algorithms, Advanced Computer Networks, Network Security, Cryptology, Topics in Cryptanalysis

POSITIONS OF RESPONSIBILITY

Teaching Assistant -- Programming & Data Structures, IIT Hyderabad

Mentor students in **problem-solving, debugging and solution validation** across programming/data-structures labs; support assessments and grading.

Technical Mentor -- New Hires & Interns

Mentored **2 new hires and 3 interns** on project architecture, analytical problem-solving and end-to-end data-issue investigation.

Teaching Assistant -- ACM India Summer School on Symmetric Key Cryptography, IIT Hyderabad: Supported PRG/PRF, DES/AES and Differential Cryptanalysis theory/lab sessions.

Volunteer Mentor -- Linux & Systems Security Bootcamp, IIT Hyderabad: Assisted with Linux configuration, lab troubleshooting and quizzes.

Sunshine PG Mentor -- IIT Hyderabad

First point of contact for **25 PG mentees**, connecting students with appropriate counselling/resources when required.

HOBBIES & INTERESTS

Volleyball | Sanskrit | Indian philosophy | Fitness & Mobility